

The N400 effect in children: Relationships with comprehension, vocabulary and decoding

Lisa M. Henderson^{a,*}, Heidi A. Baseler^a, Paula J. Clarke^b, Sarah Watson^a, Margaret J. Snowling^a

^a Department of Psychology, University of York, United Kingdom

^b School of Education, University of Leeds, United Kingdom

ARTICLE INFO

Article history:

Accepted 11 December 2010

Available online 26 January 2011

Keywords:

N400

Event-related potentials

Semantic processing

Vocabulary

Listening comprehension

Decoding

ABSTRACT

Using event-related potentials (ERPs), we investigated the N400 (an ERP component that occurs in response to meaningful stimuli) in children aged 8–10 years old and examined relationships between the N400 and individual differences in listening comprehension, word recognition and non-word decoding. Moreover, we tested the claim that the N400 effect provides a valuable indicator of behavioural vocabulary knowledge. Eighteen children were presented with picture-word pairs that were either 'congruent' (the picture depicted the spoken word) or 'incongruent' (they were unrelated). Three peaks were observed in the ERP waveform triggered to the onset of the picture-word stimuli: an N100 in fronto-central channels, an N200 in central-parietal channels and an N400 in frontal, central and parietal channels. In contrast to the N100 peak, the N200 and N400 peaks were sensitive to semantic incongruity with greater peak amplitudes for incongruent than congruent conditions. The incongruity effects for each peak correlated positively with listening comprehension but when the peak amplitudes were averaged across congruent/incongruent conditions they correlated positively with non-word decoding. These findings provide neurophysiological support for the position that sensitivity to semantic context (reflected in the N400 effect) is crucial for comprehension whereas phonological decoding skill relates to more general processing differences reflected in the ERP waveform. There were no correlations between ERP and behavioural measures of expressive or receptive vocabulary knowledge for the same items, suggesting that the N400 effect may not be a reliable estimate of vocabulary knowledge in children aged 8–10 years.

© 2011 Elsevier Inc. All rights reserved.

1. Introduction

Language comprehension is a highly complex process that fundamentally relies upon the efficient retrieval of word meaning (Kintsch & Rawson, 2005; Perfetti, 2007). Many behavioural studies have shown that individual differences in comprehension skill relate to performance on a variety of tasks that require access to word meaning (e.g., meaning judgement and providing definitions) (e.g., Nation, Marshall, & Snowling, 2001; Nation & Snowling, 1998). However, the responses in such 'off-line' tasks provide a measure of the end point of semantic processing (after the semantic information has been retrieved) rather than measuring semantic processing as it unfolds in real time. Moreover, there is a lack of neurophysiological support for the view that access to word meaning relates to individual differences in comprehension skill, particularly in children. The present study used EEG (electroencephalography) as an 'on-line' measure of semantic processing and investigated the event-related potentials (ERPs) that occur in

response to picture-word pairs that semantically match or mismatch. We explored whether aspects of these ERPs relate more strongly comprehension than to decoding of print in children aged 8–10 years. Furthermore, we examined whether these ERPs can provide a reliable and valid index of children's behavioural vocabulary knowledge (as measured by expressive and receptive tests) for the same words.

ERP studies have been critically important for the study of language since they provide a continuous temporal record of brain activity from the beginning of stimulus onset without interference from overt decision or response processes that are often characteristic of 'off-line' behavioural tasks (Osterhout, 2000; Osterhout, McLaughlin, & Bersick, 1997). ERPs are obtained by averaging responses to stimuli in different conditions, yielding a distinctive succession of positive and negative deflections (or peaks) which are labelled by their polarity (P = positive, N = negative) and order of occurrence (e.g., N1, N2) or point of occurrence post-stimulus onset (e.g., N200, P300). An ERP recording consists of multiple discrete components or brain waves that are related to different stages of stimulus processing. These components usually arise from different scalp locations. For instance, ERP components associated with auditory processing may be observed in temporal

* Corresponding author.

E-mail address: L.Henderson@psych.york.ac.uk (L.M. Henderson).

sites, visual processing in occipital sites, and semantic processing in central and parietal sites on the scalp.

The ERP component of focus in the present study is the N400. The N400 is an ERP response to the presentation of words, pictures and other meaningful stimuli which appears as a negative-going deflection in the ERP waveform in adults (Brown & Hagoort, 1993; Federmeier, 2007; Hagoort, 2008; Lau, Almeida, Hines, & Poeppel, 2009) and in infants and children (Coch, Maron, Wolf, & Holcomb, 2002; Friedrich & Friederici, 2004, 2006; Hahne, Eckstein, & Friederici, 2004; Holcomb, Coffey, & Neville, 1992) (for reviews see Kutas & Federmeier, 2000; Lau, Phillips, & Poeppel, 2008). The N400 typically occurs in centro-parietal channels and onsets between 200 and 300 ms reaching its peak amplitude at around 400 ms post-stimulus onset. Several standard psycholinguistic paradigms have been used to show that the N400 response to words depends on contextual information. For instance, in the picture-word paradigm a spoken or written word is paired with a picture that is either congruent (semantically related) or incongruent (semantically unrelated). When the amplitude of the N400 in response to the target stimulus is more negative for incongruent than congruent conditions this is termed the 'N400 effect'. Some have argued that the N400 effect reflects the impact of context on semantic access (Federmeier, 2007; Kutas & Federmeier, 2000; Lau et al., 2009) while others have proposed that it reflects a post-access attempt to integrate semantic information into context (Brown & Hagoort, 1993; Hagoort, 2008; Osterhout & Holcomb, 1992). Many studies using the picture-word paradigm have reported N400 effects in infants (Friedrich & Friederici, 2004, 2005, 2006; Torkildsen et al., 2006; Torkildsen et al., 2008); however, there are fewer studies that report the N400 effect in school-aged children (Byrne et al., 1999; McCleery et al., 2010).

Previous studies have shown that the N400 effect is sensitive to individual differences in comprehension skill in infants (Friedrich & Friederici, 2006) and adults (Landi & Perfetti, 2007), although this remains to be investigated in middle childhood. This finding supports the lexical quality hypothesis (Perfetti, 2007) which proposes that variation in the quality of lexical representations has direct consequences for word recognition and text and discourse comprehension. High lexical quality is defined as well-specified orthographic and phonological representations of words and clearly defined but flexible semantic representations of meaning, allowing for quick and reliable meaning retrieval. More specifically, Perfetti (2007) proposes that variation in the breadth, depth and speed of retrieval of word meanings is particularly important in determining the extent to which an individual achieves a good understanding of language. In support of this position, an accumulation of behavioural studies suggest that children with comprehension difficulties show poorer performance on tasks that require access to word meaning but not word form, when they are compared to decoding matched controls (e.g., Nation & Snowling, 1998, 1999; Nation et al., 2001).

To provide neurophysiological evidence for the relationship between semantic processing and comprehension skill, Landi and Perfetti (2007) investigated the ERPs that occur in response to semantic and phonological tasks in adults with good levels of decoding skill but poor reading comprehension (less-skilled comprehenders) and adults with good levels of decoding and reading comprehension (skilled comprehenders). In the 'semantic tasks' participants saw a series of two pictures (picture–picture) or a series of two words (word–word) and were asked to decide if they were semantically related. In the 'phonological task' they heard two words (e.g., bear–bare) and were asked to decide if they sounded the same. For all tasks an early positive going waveform was observed that began around 150 ms and continued until 250 ms (P200) and a negative-going waveform was observed at around 400 ms (N400). Both P200 and N400 components were

sensitive to differences between skilled and less-skilled comprehenders during the semantic (but not the phonological) task. These results suggest that individual differences in reading comprehension skill relate to neurophysiological markers of semantic but not phonological processing. However, Landi and Perfetti focussed on adults and it remains to be determined whether neurophysiological markers of semantic processing relate to individual differences in comprehension skill in children.

The skilled and less-skilled comprehenders in Landi and Perfetti (2007) were selected to vary on comprehension and not decoding. Hence, it remains unclear whether individual differences in the ability to decode print also relate to the ERPs that occur when participants respond to matching/mismatching word/picture stimuli, or rather whether the N400 effect is more specifically related to comprehension. It is well established that word decoding and listening comprehension each build on different oral language skills; decoding is underpinned by phonological skills whereas listening comprehension depends on semantic and grammatical skills (Bishop & Snowling, 2004). Therefore, if the N400 effect captures an aspect of semantic processing then it should relate more strongly to comprehension than to decoding. The present study tests this hypothesis; if it is supported it will increase current understanding of the functional significance of the N400, by suggesting that it captures a process important for extracting meaning from language (i.e., comprehension), rather than it being related to word recognition more generally.

The absolute amplitude of ERP waveforms in response to word and picture stimuli has also been related to individual differences in reading skills. For instance, in Landi and Perfetti (2007), regardless of the experimental condition, the skilled comprehenders showed positive deflections with greater absolute amplitude than the less-skilled comprehenders whereas the less-skilled comprehenders had more negative waveforms. They proposed that "this general difference in amplitude could reflect an underlying processing difference" (p. 42). Similar findings have been reported by Meyler and Breznitz (2005) who found that individuals with dyslexia showed negative deflections with greater absolute amplitude than controls. Landi and Perfetti (2007) also highlighted a number of previous ERP studies in which less-skilled readers have generally shown more negative ERP waveforms than skilled readers (Perfetti, Wlotko, & Hart, 2005; Segalowitz, Wagner, & Menna, 1992; Yang, Perfetti, & Schmalhofer, 2005). A more exploratory aim of the present study is to further examine these findings, and in particular, whether differences in the averaged amplitude of the ERP waveforms (averaged across experimental conditions) are associated with decoding skills, comprehension or both.

A number of studies have suggested that ERPs may have clinical use in the measurement of vocabulary knowledge in individuals who have difficulties with the demands of conventional vocabulary tasks (e.g., making expressive responses and semantic decisions, and/or pointing to stimuli etc.). These studies have shown that standardised measures of receptive vocabulary knowledge can be adapted for computer presentation and simultaneous ERP recording (Byrne, Dywam, & Connolly, 1995a, 1995b; Connolly, Byrne, & Dywan, 1996; Marchand, D'Arcy & Connolly, 2002). Byrne et al. (1999) investigated the use of ERP to assess age-appropriate receptive vocabulary in typically developing children aged 5–6, 7–8, 9–10, and 11–12 years. The task used an adaptation of the Peabody Picture Vocabulary Test (PPVT-R; Dunn & Dunn, 1981); in this passive computerised version children saw a picture and heard a spoken congruent or incongruent word (the correct or incorrect word for the picture). As predicted, the amplitude of the N400 was significantly greater for the incongruent picture-word pairs than for the congruent picture-word pairs for all age groups. However, there were developmental changes in the topography of the N400 effect; 5–6 year-olds showed a broadly distributed N400 in

frontal, centroparietal and temporal channels that became increasingly concentrated in centro-parietal channels with age. This was interpreted as reflecting the fine-tuning of information-processing strategies across development. Unfortunately, Byrne et al. did not report correlations between the amplitude or latency of the N400 and behavioural measures of vocabulary knowledge. Therefore, an outstanding question that is examined in the present study is whether behavioural vocabulary knowledge of a given set of items relates to quantitative features of the N400 (i.e., peak amplitude and peak latency) for the same items. There is a general need in this field to combine behavioural and neurological techniques to enable a better understanding of whether we are measuring the same underlying processes with each methodology.

The ERP task that was used in the present study was adapted from a standardised measure of receptive vocabulary (the British Picture Vocabulary Scales, 3rd Edition; BPVS-III; Dunn & Dunn, 2009). Therefore, to investigate relationships between ERP performance and behavioural performance, we included a behavioural measure of receptive vocabulary involving the same stimuli as in the ERP task that involved children selecting one of four pictures to match a spoken word. In addition, we included a behavioural measure of expressive vocabulary that involved the children providing definitions of the same words that were used in the ERP task. Whilst the receptive vocabulary task only provided a shallow indication of familiarity with the item set, the expressive task was included to investigate whether ERPs can also provide an accurate measure of the depth of a child's vocabulary knowledge.

1.1. Aims and hypotheses of the current study

The present study had four main goals. First, we aimed to provide much needed evidence of the N400 effect in school-aged children. Based on Landi and Perfetti (2007), we hypothesised that deflections would occur in the waveform at approximately 100–200 ms post-stimulus onset and at approximately 400 ms post-stimulus onset in response to the word-picture pair. More specifically, we predicted that the peak amplitude of the ERP waveform at approximately 400 ms post-stimulus onset would be more negative for semantically incongruent picture-word pairs than for semantically congruent picture-word pairs (the N400 effect). Specific hypotheses were not made for the deflection at 100–200 ms, since this was not the focus of the present study.

Second, we aimed to provide neurophysiological support for the importance of access to word meaning and sensitivity to semantic context to language comprehension in children. To this end, we hypothesised that the magnitude of the N400 effect would show a stronger and more positive correlation with measures of listening comprehension than of word recognition or non-word decoding skills. We also aimed to provide further information on the functional significance of the N400 effect by examining whether it reflects a process important for comprehension, rather than the earlier stages of word or stimulus recognition.

Thirdly, in light of previous observations that the absolute amplitude of ERP waveforms relates to reading skills (e.g., Landi & Perfetti, 2007; Meyler & Breznitz, 2005; Perfetti et al., 2005; Segalowitz et al., 1992; Yang et al., 2005), we tested a more exploratory hypothesis that the amplitude of negative ERP deflections when averaged across congruent and incongruent conditions would be positively correlated with word recognition, non-word decoding skills and/or comprehension. More specifically, we predicted that better reading/comprehension skills would be associated with lower absolute amplitude of the negative deflections.

Finally, we aimed to provide further information on the extent to which ERP indicators of vocabulary knowledge are predictive of behavioural performance for the same items. We hypothesised that the magnitude of participants' N400 effect would be inversely

correlated with their vocabulary knowledge of the experimental items on behavioural measures of receptive and expressive vocabulary.

2. Method

2.1. Participants

Children aged 8–10 years were the focus of this study since at this age listening comprehension can be reliably measured and is a strong predictor of reading comprehension (Nation & Snowling, 1997). Furthermore, this age group were deemed to be more likely to sit comfortably and concentrate throughout the EEG and behavioural testing, thus providing a more reliable assessment of whether ERP and behavioural measures of vocabulary knowledge are correlated. The age range of 8–10 years was considered sufficiently narrow to ensure the latency of the ERP peaks would not differ significantly across participants (Hahne et al., 2004; Picton et al., 2000). Eighteen children aged 8–10 years were recruited for this study (8 males; 10 females) from six primary schools in York, UK. Parents reported that all children were right-handed. The Edinburgh Handedness Questionnaire (adapted from Oldfield, 1971) confirmed that 16 children were right-handed and two were ambidextrous. All children were in good health, none of them had received a head injury and they had normal or corrected-to-normal vision and hearing. This research was approved by the York Neuroimaging Centre Research and Ethics Committee. Informed consent was obtained from the parents and the children.

2.2. Design and materials

2.2.1. Background measures

Listening comprehension was measured using the Listening to Paragraphs subtest from the Clinical Evaluation of Language Fundamentals (3rd Edition; CELF-III, Semel, Wiig, & Secord, 2000). Listening comprehension was measured rather than reading comprehension to obtain a purer measure of comprehension that is not confounded by word-level reading skills. To ensure a wide enough range of stories were administered, the children listened to six stories of increasing difficulty (rather than two as recommended in the manual) and answered open-ended comprehension questions after each. The comprehension questions tapped understanding of the main themes, detail, sequencing of events, inference making and prediction. Standardised listening comprehension scores were derived from the two passages appropriate for the child's age. To carry out a full assessment of children's word-level reading skills, both word recognition and non-word decoding were measured using the Test of Word Reading Efficiency (TOWRE, Torgesen, Wagner, & Rashotte, 1999). Word recognition was assessed by asking the children to read as many words as possible in 45 s from a card on which the words gradually increased in difficulty. Decoding was assessed by asking the children to read as many nonsense words as possible in 45 s. Together, the scores from the two reading tests form a total reading efficiency score. As shown in Table 1, the mean standard scores on each measure were comfortably in the average range.

2.3. EEG testing

2.3.1. ERP task stimuli

The computerised ERP task consisted of 104 coloured picture stimuli paired with either their correct picture name (congruent) or incorrect picture name (incongruent). Thus, there were 104 congruent trials and 104 incongruent trials (see Appendix A). All pictures and their congruent picture names were taken from the

Table 1Background and vocabulary measures for all children ($n = 18$).

	Mean	SD	Range
Age (years)	10.21	0.83	8.58–10.92
Listening comprehension ^a	9.06	2.18	4–12
Word reading ^b	109.33	11.10	85–127
Non-word decoding ^b	110.00	13.38	80–129
Total word reading ^b	111.61	13.94	79–134
Receptive vocabulary (out of 104)	96.5	8.18	75–104
Definitions (out of 312)	174.78	32.01	109–215

^a Scaled scores (0–19, 8–12 average range).^b Standard scores (85–115 = average range).

BPVS-III (with the publishers permission; Dunn & Dunn, 2009). Ninety-two pictures were taken from the standardised Year 5 set (for 9–10 year olds) and 12 pictures were included from the Year 6 set (for 10–11 year olds) to ensure that the list contained items of varying difficulty. The incongruent words were selected from the Children's Printed Word Database (<http://www.essex.ac.uk/psychology/cpwd>) and were matched to the congruent words on initial letter-sound, syllable length, and frequency. The MRC Psycholinguistic Database (Wilson, 1988) was used to match the congruent and incongruent word lists for familiarity, imageability and concreteness. Twenty-four undergraduate students were asked to rate the 104 congruent and incongruent picture–word pairs for relatedness on a seven point scale (1 = unrelated to each other; 7 = related to each other). The mean rating for the congruent pairs was significantly higher (mean = 6.34, SD = 0.77) than the mean rating for the incongruent pairs (mean = 1.46, SD = 0.56), $t(103) = 55.57$, $p < .001$. The words were digitally recorded in a sound attenuated booth by a female speaker of British English.

2.3.2. ERP task design

The experiment was administered using Presentation (Version 9.9, Neurobehavioural Systems, Inc., Albany, CA, USA). Each trial began with a central fixation cross displayed in the centre of a screen for 2000 ms (the inter-trial interval). A high-resolution projector screen was used at a viewing distance of 150 cm which displayed a white background throughout (size of projected white background 40 × 30 cm; widest dimension of projected picture stimuli 10–15 cm). Following the fixation cross, a coloured picture was presented in the centre of the screen and simultaneously a spoken word was presented binaurally through earphones at 55 dB (SOA 0 ms). The picture remained on screen for a total of 2000 ms to ensure that ERP offset responses to picture termination did not interfere with the recording of auditory ERPs. A trigger was placed at the onset of each word stimulus. Two versions of the experimental list were produced that were pseudo-randomised for congruency and difficulty, with the stipulation that trials containing the same picture were separated by at least 10 trials. Since each child saw each picture twice during the course of the EEG experiment all children were pre-exposed to the pictures (see Pre-Exposure/Naming task below) to reduce repetition and novelty effects to which several ERP components are sensitive (Friedman, Cycowicz, & Gaeta, 2001).

One advantage of using ERPs is that behavioural tasks (e.g., meaning judgements), that do not occur in natural language processing, do not have to be used. A drawback of these tasks is that they elicit decision-related P300s, which may obscure ERP components of interest (Kutas & Van Petten, 1988; Picton et al., 2000). Thus, we chose to use a passive task rather than recording behavioural responses at the same time as the ERPs. To ensure that children were attentive throughout the task 10 catch trials were randomly interspersed in the experimental list that consisted of cartoon characters for which children had to press a response

button. Seven practice items were administered (three congruent, three incongruent, one catch trial).

2.3.3. EEG recordings

ERPs time-locked to stimulus onset were recorded from silver-silver chloride electrodes at 25 locations in accordance with the 10–20 System (AFz, Fz, FCz, Cz, CPz, Pz, Oz, AF3, AF4, F3, F4, F7, F8, FC1, FC2, FC3, FC4, C3, C4, P3, P4, PO7, PO8, TP1 and TP2) and at two lateral temporal locations (LT1, LT2) placed laterally to Cz at 33% of the interaural distance (following Hald, Steenbeek-Planting, and Hagoort (2007)).

We used linked mastoids as a reference to facilitate the comparison of similar data from other laboratories; this was the method adopted by studies that the present study was attempting to replicate and extend (e.g., Byrne et al., 1999; Marchand, D'Arcy, & Connelly, 2002) and has been commonly used by other developmental studies in the field (e.g., Atchley & Kwasny, 2003; Conboy & Mills, 2006; Holcomb et al., 1992; McCleery et al., 2010; Nanova, Lyamova, Hadjigeorgieva, Kolev, & Yordanova, 2008; Torkildsen et al., 2008). Picton et al. (2000) highlighted the controversies surrounding the use of linked mastoids and recommended using an average reference; however, it has been argued that the average reference should only be used when a large number of channels have been recorded with good coverage of the scalp (Luck, 2005) and in this study a restricted number of channels were specifically selected to measure the N400. Furthermore, Conboy and Mills (2006) report the results of an ERP pilot study using auditory stimuli that examined possible distortions in the distribution of scalp activity resulting from forced mastoid linkage. This was determined by recording from one mastoid, using the other as a reference, and linking the mastoids off-line. These data were compared with data recorded using linked mastoids and no significant differences were found.

FPZ was the ground electrode. Electrodes were attached to an elastic electrode cap (EASYPAP, Herrsching, Germany). Electrooculograms of vertical and horizontal eye movements were recorded with bipolar electrodes placed vertically and horizontally around the right eye. Impedances were kept below a threshold of 20 k Ω but were typically below 5 k Ω . EEG data were acquired through a 4D Neuroimaging integrated MEG/EEG system (San Diego, CA, USA), using 32-channel SynAmps II amplifiers. Analogue signals were digitized at a rate of 678.17 Hz band pass filtered (.01–100 Hz) and amplified 1000 times.

EEG signals were analysed using EEGLAB (<http://sccn.ucsd.edu/eeqlab/>), a publicly available analysis toolbox written for MATLAB (The Mathworks, Natick, MA, USA). Trials containing eye blink, eye movement or channel artefact were removed via visual inspection. The mean proportion of trials removed across participants was .13 for incongruent trials (SD = .06; range .08–.28) and .14 for congruent trials (SD = .08; range .03–.30). Data were segmented into 1500 ms epochs starting 500 ms before the onset of the word. The epochs were baseline-corrected by subtracting the mean amplitude between –500 and 0 ms. Individual ERP waveforms, time-locked to the onset of the word, were generated for each child, condition and channel. A 30 Hz low pass filter was applied to all of the resulting average waveforms.

2.4. Behavioural testing

2.4.1. Pre-exposure/naming

Prior to the ERP task children were pre-exposed to all picture stimuli. In this task children were presented with each picture in turn using Eprime (Schneider, Eschman, & Zuccolotto, 2002) (Psychology Software Tools, Inc., <http://www.pstnet.com/eprime.cfm>). To ensure that children were attending to each picture they were asked to name them (no feedback was given so as not to interfere

with performance on the ERP/behavioural tasks). Picture naming accuracy was recorded by the examiner and naming RT was recorded with a voice key through Eprime. Mean picture naming RT for correct items was 1578 ms ($SD = 575$ ms; 880–2717 ms). Although the children tended to name the pictures using semantically appropriate labels (mean correct 91.22%; $SD = 4.01\%$; range 79–100%), they were less likely to provide the precise name supplied by the BPVS-III (naming agreement 19.44%; $SD = 3.85\%$; range 14–30%). For example, for the picture depicting “links”, many children said “chain” or “chain of links”. We do not believe that the low convergence of naming is a limitation of the present study, since the logic of the ERP task depends on the children’s ability to detect whether picture–word pairs are semantically related or unrelated. Indeed, low convergence of naming may indicate that the participants varied in the depth of their understanding of the target words which is important for the investigation of whether individual differences in vocabulary knowledge correlate with properties of the ERP responses. Moreover, the participants made few errors on the receptive vocabulary task where they decided which picture out of four best matched the target word which was spoken by the examiner (described below).

2.4.2. Receptive vocabulary

A multiple choice task was administered after the EEG recording to assess children’s receptive understanding of the experimental items. In this task, each picture was presented in an array of four pictures and the children had to choose the one picture that matched the word spoken by the examiner. The words and pictures (including the distracters) were the same ones that were used in the ERP task. All distracters were unrelated to the target word but were within the same word class (noun, verb or adjective). One point was awarded for a correct answer. As shown in Table 1, the errors on this task were low overall, with the majority of children being familiar with most items. Thus, although naming agreement of the pictures was low, the children were able to match the majority of pictures to their names.

2.4.3. Definitions

To measure the depth of expressive vocabulary knowledge of the experimental items participants defined all of the 104 items after the EEG recording. In this task, each word was spoken by the examiner and the children were asked “What does X mean?” or “What is/are X?”. No picture cues were available. A scoring system was developed for each item so that each definition was graded from 0 to 3. Two researchers scored the children’s definitions independently and inter-rater reliability was high (mean correlation coefficient across all items $r = .91$, $p < .001$; range $r = .59$ – 1.0). Table 1 shows the mean total score for the participants on this task. Children’s scores on this task were significantly correlated with scores on the receptive vocabulary task, $r = .65$, $p < .01$, and familiar items on the receptive vocabulary task were associated with higher scores on the definitions task than unfamiliar items, $t(17) = 11.14$, $p < .001$.

2.5. Procedure

Children were first pre-exposed to the picture stimuli in the picture naming task. This took approximately five minutes. Whilst the EEG cap was fitted, children watched an age-appropriate DVD to occupy their attention and facilitate electrode placement. It was ensured that the CZ electrode was positioned at the midpoint between the nasion and inion and left and right pre-auricular notches. The scalp electrode sites were prepared using a sodium based electrolyte abrasive gel. The mastoid channels were prepared first and were applied directly to the skin, followed by the scalp channels and finally the VEOG and HEOG eye channels.

Before seating the child in the electromagnetically shielded room impedances were tested using an impedance metre (SIGGI II, EASYCAP, Herrsching, Germany) and they were re-checked in the shielded room directly before testing using NeuroScan SCAN 4.0 software. The children were seated 150 cm in front of the projector screen and earphones (model ER30, Etymotic Research Inc., Elk Grove Village, IL, USA) were fitted. Parents and examiners remained outside the room; infrared video camera and intercom systems were used to observe and speak to the child. Practice stimuli were presented to the children to familiarise them with the catch trials and to ensure they could hear the words and see the pictures clearly. Catch trials comprised 5% of the stimulus set and consisted of pictures of cartoon characters with no accompanying word. The EEG data acquisition period lasted approximately 25 min, including three breaks. The children were asked to stay as still as possible during the recording sessions. They were instructed to watch the pictures and listen to the words and to press a response button as quickly as possible using their right index finger on a response box (Lumitouch, Photon Control Inc.) placed on their lap whenever they saw a cartoon character.

On completion of the ERP task, the cap was removed and the child had a break before the behavioural testing began. During this assessment (which lasted approximately one hour) the standardised measures and bespoke measures of vocabulary were administered in a pre-determined order for each child. The definition task was split up into three blocks that were interleaved with the listening comprehension task (i.e., two stories were administered between each block of definitions), followed by the receptive vocabulary task which was split up into two blocks that were interleaved between the word reading and non-word reading tasks. Finally, the handedness questionnaire was administered.

3. Results

Participant’s data were retained if they scored above 80% correct on the catch trials (mean percent catch trials correct 97.22%, $SD = 6.69$, range 80–100%). The mean percent catch trials correct is based on 17 out of the 18 participants as the catch-trial (but not EEG) data were lost for one participant due to a computer malfunction (the experimenter monitored this participants’ catch trial responses to ensure they responded correctly).

Grand average and individual subject data were inspected prior to any statistical analysis. The EEG epochs related to congruent and incongruent conditions were sorted by computer, and epochs for each channel and condition were averaged. Between 50 and 150 ms post-stimulus onset, there was a small and early negative deflection that was most prominent in frontal channels but was widely distributed (N100; see Figs. 1–3). Following this, there was another negative deflection between 150 and 250 ms post-stimulus onset, particularly at centro-parietal channels, when differences between congruent and incongruent conditions began to emerge (N200; see Figs. 1 and 4). Between 300 and 500 ms, a negative peak was observed most prominently in frontal and central channels for both conditions but was more negative for the incongruent condition (N400; see Fig. 1). Inspection of the topographic maps (Fig. 2) reveals that the N200 may reflect the onset of the widely reported N400. The greater negativity for the incongruent than congruent condition persists for the remainder of the epoch.

For each peak (i.e., N100, N200, N400), the peak amplitude and peak latency for congruent and incongruent conditions was calculated separately for each child and for each channel. Hence, the amplitude and latency at single time points (the peaks) were analysed rather than time windows. Electrodes were selected for further statistical analysis if at least 17 out of the 18 children in the sample showed the peak of interest. To examine whether the

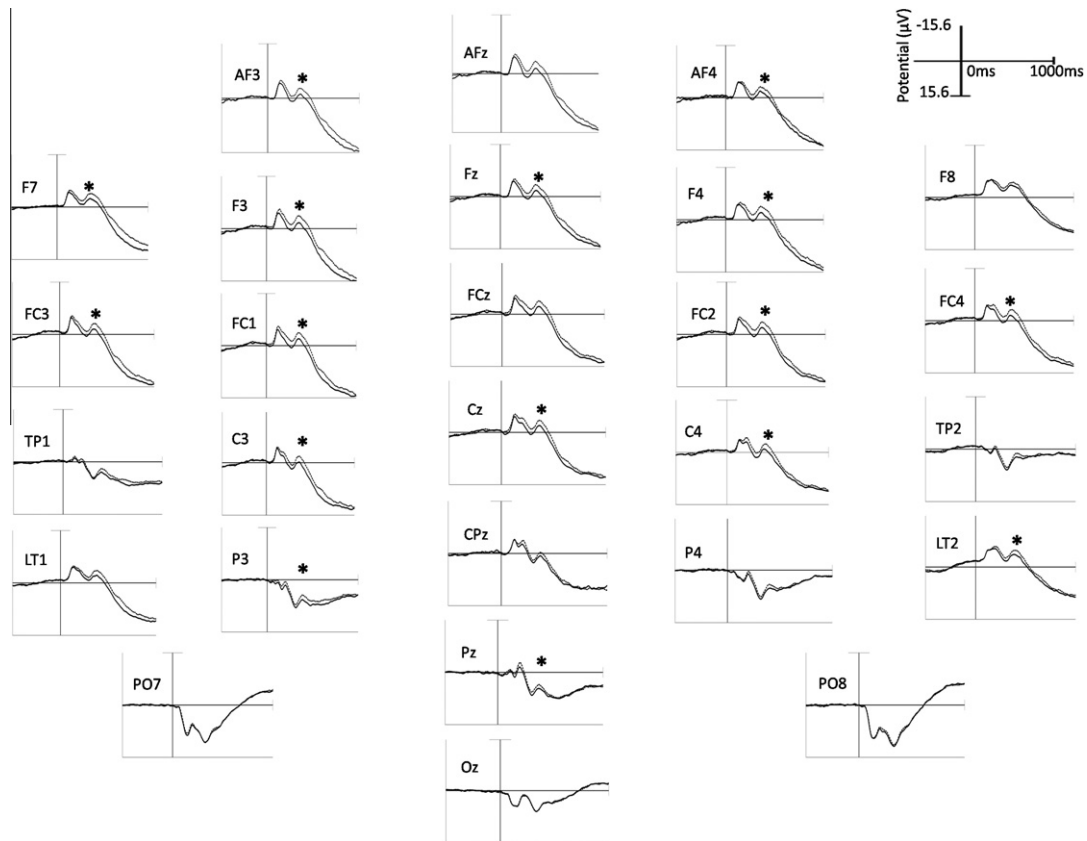


Fig. 1. ERP waveforms for all channels for the congruent condition (solid line) and the incongruent condition (dotted line). Note. Pairwise *t* tests were performed to test the significance of the N400 incongruity effect for each channel using Bonferroni correction (* $p < .00185$). The degrees of freedom for each comparison typically exceeded 14, with the exception of Oz ($df = 9$), PO7 ($df = 5$), PO8 ($df = 2$), TP1 ($df = 13$), TP2 ($df = 12$); *t* tests were not performed for these channels.

amplitudes for each peak were significantly more negative for incongruent than congruent conditions, peak amplitudes for the N100, N200 and N400 were entered into separate repeated measures ANOVAs with Condition (congruent, incongruent) and Channel as within-subjects variables. For each ANOVA, an alpha value of .05 was used for main effects and interactions and Bonferroni correction was applied when breaking down main effects or interactions when there were more than two comparisons. As recommended by Picton et al. (2000), Greenhouse-Geisser correction was used where the assumption of sphericity was violated. Spearman's rho correlations were then performed between the peak latency and the congruency effects (difference between congruent and incongruent conditions, termed the 'N100/N200/N400 effect', etc.) for each peak and the background measures (listening comprehension, decoding skills, vocabulary measures). When present in each individual, congruent and incongruent peaks were always of the same polarity. Therefore, congruent and incongruent peak amplitudes were averaged for each peak in order to perform Spearman's rho correlations between the averaged amplitudes and background measures. These correlations enabled us to evaluate the hypothesis that overall amplitude of ERPs relates to decoding and/or comprehension skill. For all correlations, an alpha value of .01 was used in accordance with Bonferroni correction to control for the multiple comparisons.

3.1. N100

An N100 peak was present for 17 out of 18 children in channels AFz, Fz, FCz, FC1 and FC2; these channels were therefore selected for further analysis of the congruency effect. The mean peak la-

tency across these channels was 131.96 ms (SD = 10.35 ms; range 113.12–149.44 ms) (see Fig. 3). There was no correlation between chronological age and the latency of the N100, $r(16) = .14$, $p > .05$; hence statistical analysis was performed for all 17 children combined.

A repeated measures ANOVA was performed with Condition (congruent, incongruent) and Channel (AFz, Fz, FCz, FC1, FC2) as the within-subjects variables. The main effect of Condition did not reach significance ($F(1, 15) = 3.63$, $p = .08$, $\eta_p^2 = .20$; amplitudes were statistically similar for the congruent and incongruent conditions (see Fig. 2). There was a significant main effect of Channel ($F(2.13, 31.98) = 4.77$, $p < .05$, $\epsilon = .53$, $\eta_p^2 = .24$). The peak amplitude was greater for FC1 (mean -5.5 mV; SD = 2.5 mV) and FCz (mean -6 mV; SD = 3 mV) channels than for AFz (mean -4.5 mV; SD = 2.5 mV; $p < .001$) and FC2 (mean -4.5 mV; SD = 2.5 mV, $p < .001$). Thus, the N100 was maximal in fronto-central midline and LH channels (see Fig. 2). The Condition $\times$ Channel interaction was not significant ($F(1.67, 24.98) = 0.33$, $p > .05$, $\epsilon = .42$, $\eta_p^2 = .02$).

There were no significant correlations between the peak latency of the N100 (for channels AFz, Fz, FCz, FC1 and FC2 or combined) and the background measures. There was a marginally significant ($p < .05$) correlation between peak amplitude (averaged across congruent and incongruent conditions) at FCz and phonemic decoding; better decoders showed lower absolute N100 amplitudes overall (Table 2). There was also a strong positive correlation between the magnitude of the N100 effect at AFz and listening comprehension (Table 3); better comprehenders showed greater differences in amplitude between congruent and incongruent conditions 100 ms post-stimulus onset.

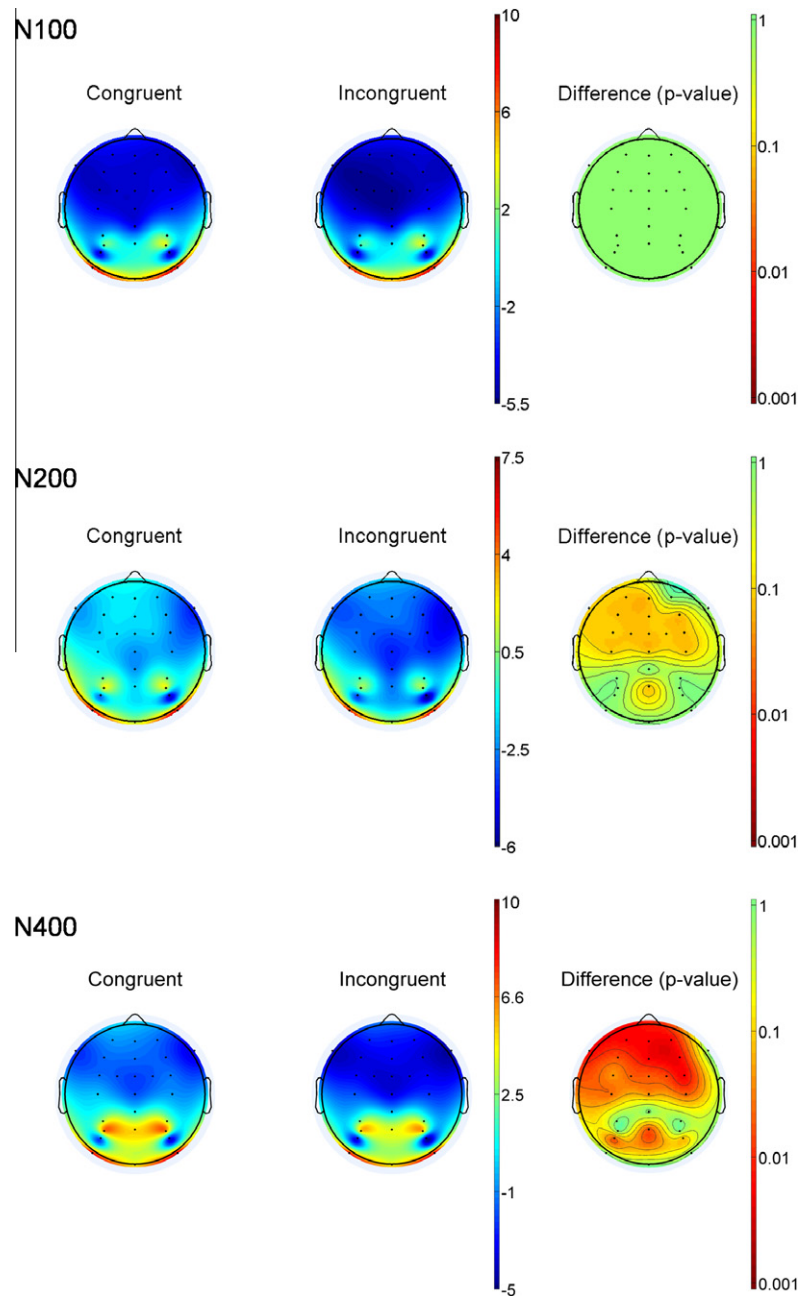


Fig. 2. Topographic maps for the N100 (4a), N200 (4b) and N400 (4c) mean to peak latencies (132 ms, 208 ms and 375 ms from stimulus onset, respectively) for congruent and incongruent conditions and the difference between the two conditions (represented as a *p* value, controlling for multiple comparisons).

3.2. N200

N200 peaks occurred for the majority of participants in channels Cz ($n = 17/18$ participants), CPz ($n = 18/18$), Pz ($n = 18/18$), C4 ($n = 18/18$), and P4 ($n = 18/18$) and these channels were therefore selected for further analysis. Mean peak latency averaged across these channels was 208.15 ms (SD = 19.88 ms; range 170.62–246.12 ms) (see Figs. 1 and 3). Age and peak latency were not significantly correlated, $r = .09$, $p > .05$, hence statistical analysis was performed for the group as a whole.

A repeated measures ANOVA was performed with Condition (congruent, incongruent) and Channel (Cz, CPz, Pz, C4, P4) as repeated measures variables and peak amplitude as the dependent variable. In contrast to the N100, the main effect of Condition reached significance ($F(1, 16) = 12.06$, $p < .01$, $\eta_p^2 = .43$; the ampli-

tude was greater for the incongruent than for the congruent condition see Fig. 2). There was also a significant main effect of Channel ($F(2.38, 38.05) = 18.07$, $p < .001$, $\epsilon = .60$, $\eta_p^2 = .53$); P4 (0.1 mV; SD = 3.5 mV) had lower absolute negative amplitude than Cz (−4.5 mV; SD = 3.5 mV), $t(16) = -5.46$, $p < .001$, CPz (−4 mV; SD = 3.5 mV), $t(17) = -7.07$, $p < .001$, Pz (−3 mV; SD = 3 mV) $t(17) = -6.14$, $p < .001$, and C4 (−3 mV; SD = 3 mV) $t(17) = -5.25$, $p < .001$. The Channel $\times$ Condition interaction was not significant ($F(1.96, 31.40) = 0.38$, $\epsilon = .49$, $p > .05$, $\eta_p^2 = .02$). Therefore, the N200 was greater for incongruent than congruent trials and was most prominent in central, centro-parietal and parietal midline channels. Fig. 4 shows the N200 peak at Cz and CPz.

There were no significant correlations between the peak latency for the N200 and any of the background measures. However, there were significant positive correlations between peak amplitude

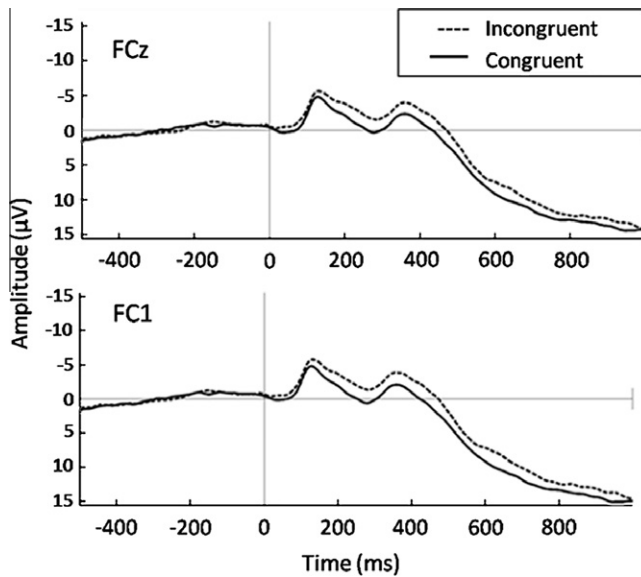


Fig. 3. N100 in channels FCz and FC1.

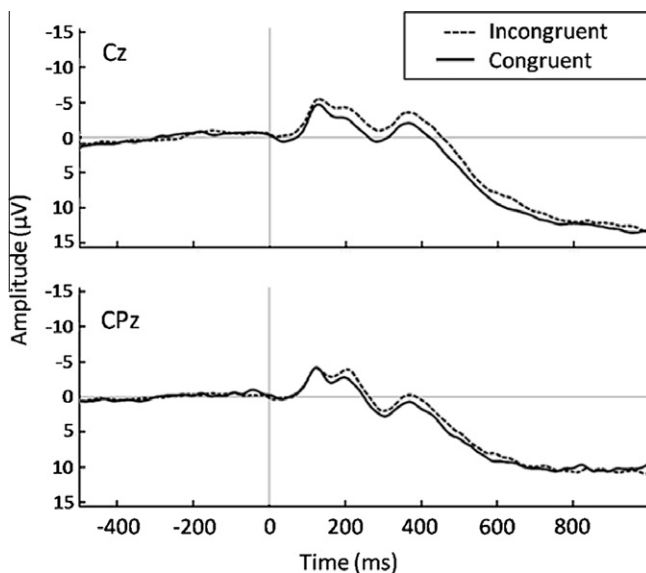


Fig. 4. N200 in channels Cz and CPz.

(averaged across congruent and incongruent conditions) at Cz and CPz and phonemic decoding (Table 2). As for the N100 amplitude, better decoders showed smaller absolute N200 amplitudes. There was also a significant positive correlation between the N200 incongruency effect and listening comprehension for Cz with better comprehenders showing a larger incongruency effect (Table 3).

3.3. N400

The N400 peak was widely distributed topographically but occurred for all children in the following midline and RH fronto-central, central, frontal and centro-parietal channels: AFz, FCz, FC1, FC2, FC3, FC4, Cz, C3, C4, F3, F4, CPz (see Fig. 1). These channels were grouped according to location for the purpose of analysis: midline (AFz, FCz, Cz, CPz), left hemisphere (LH; FC1, FC3, C3, F3), right hemisphere (RH; FC2, FC4, C4, F4). Peak latency across these channels was 375.26 ms ($SD = 21.48$ ms; range 338.03–416.82 ms). Again, there was no significant correlation between age and N400 peak latency, $r = .10$, $p > .05$.

A repeated measures ANOVA was carried out with Congruency (congruent, incongruent) and Location (Midline, RH, LH) as repeated measures variables and the N400 peak amplitude (in microvolts) as the dependent variable. The main effect of Congruency was significant ($F(1, 16) = 23.64$, $p < .001$, $\eta_p^2 = .60$; the N400 amplitude was significantly larger for the incongruent than the congruent condition. Neither the main effect of Location ($F(1.25, 20.05) = 1.64$, $p > .05$, $\epsilon = .63$, $\eta_p^2 = .09$ nor the Congruency $\times$ Location interaction ($F(2, 32) = 0.01$, $p > .05$, $\eta_p^2 = .0$) were significant. Thus, the N400 amplitude was larger for incongruent than congruent conditions, as predicted, and was broadly distributed in frontal, central, fronto-central and centro-parietal channels (see Fig. 2).

There were no significant correlations between peak N400 latency for RH, LH or midline locations and any of the background measures. There were significant positive correlations between phonemic decoding and N400 amplitudes at midline, RH and LH channel locations and a significant positive correlation between word reading and the N400 amplitude for LH channels (Table 2); children with better decoding and word reading skills showed smaller N400 amplitudes that were hence less negative. There was also a significant positive correlation between the magnitude of the N400 effect and listening comprehension for RH channels (Table 3). However, counter to predictions, there were no significant correlations between the vocabulary measures and the N400 effect.

4. Discussion

The present study was designed to investigate the N400 effect in middle childhood and examine relationships with listening comprehension and word and non-word decoding. We also explored relationships between the absolute ERP waveform amplitude and these background measures. Finally, we aimed to provide new information on the relationships between neurophysiological and behavioural indicators of vocabulary knowledge. To fulfil these aims, we measured ERPs in children whilst they heard words that either did or did not match a simultaneously presented picture.

Three deflections in the ERP waveform were revealed. The first was an N100 in left hemisphere fronto-central channels. Given its early latency, this deflection is likely to be sensory in nature and reflect early auditory attention capture (Oray, Lu, & Dawson, 2002; Vaughan & Ritter, 1969; Vogel & Luck, 2000). Generally, the N100 is assumed to reflect selective attention to basic stimulus characteristics, initial selection for later pattern recognition, and/or intentional discrimination processing (e.g., Fonaryova Key, Dove, & Maguire, 2005; Vogel & Luck, 2000). N100 peaks occur in response to both auditory and visual stimuli, with larger deflections and shorter latency for the auditory stimuli (Fonaryova Key et al., 2005). Thus, the simultaneous presentation of visual and auditory stimuli in the present experiment likely maximised the possibility of capturing the N100.

In addition, we observed an N200 in central, centroparietal and parietal midline channels. The N200 has been reported in a number of previous studies but its functional significance remains controversial. It has been interpreted as reflecting processes involved in word recognition (e.g., Rodriguez-Fornells, Schmitt, Kutas, & Munte, 2002), comprehension of individual word meanings (e.g., Conboy & Mills, 2006), and phonological expectancy (e.g. D'Arcy, Connolly, Service, Hawco, & Houlihan, 2004). Based on the results of the present study alone it is not possible to tease apart these different interpretations.

As predicted, we also observed a broadly distributed N400 in frontal, fronto-central, central and parietal channels (see Fig. 1). Consistent with previous research (e.g., Byrne et al., 1999;

Table 2

Correlations between the background measures and baseline-to-peak amplitudes (averaged across congruent and incongruent conditions) for the N100, N200 and N400.

	Listening comp	Word reading	Non-word decoding	Receptive vocab	Definitions
<i>Averaged N100 amplitude</i>					
AFZ	-.08	.28	.34	-.23	-.21
FZ	-.12	.28	.38	-.23	-.26
FCZ	-.18	.25	.53 ^b	-.17	-.35
FC1	-.17	.26	.46	-.13	-.39
FC2	-.17	.21	.44	-.17	-.28
<i>Averaged N200 amplitude</i>					
Cz	-.07	.31	.61 ^a	-.23	-.22
CPz	.07	.38	.58 ^a	-.27	-.19
Pz	-.02	.22	.41	-.22	-.36
C4	-.02	.26	.51	-.26	-.02
P4	-.05	.13	.16	-.02	-.17
<i>Averaged N400 amplitude</i>					
Midline	-.18	.44	.58 ^a	.36	-.05
RH	-.15	.40	.63 ^a	.31	-.05
LH	-.04	.57 ^a	.62 ^a	.51	-.05

^a $p < .01$.^b $p < .05$.**Table 3**

Correlations between the background measures and the N100, N200 and N400 effects (congruent–incongruent).

	Listening comp	Word reading	Non-word decoding	Receptive Vocab	Definitions
<i>N100 effect</i>					
AFZ	.74 ^a	-.05	-.08	.45	.38
FZ	.43	-.13	-.01	.37	.29
FCZ	.17	-.30	-.19	-.04	.28
FC1	.22	-.30	-.08	.19	.27
FC2	.03	-.50	-.40	.01	.30
<i>N200 effect</i>					
Cz	.59 ^a	-.11	-.24	-.28	-.25
CPz	.27	-.06	-.20	-.39	-.44
Pz	.17	.17	.02	-.15	.15
C4	.38	-.13	-.28	-.12	-.16
P4	.54 ^b	-.003	-.21	-.12	-.05
<i>N400 effect</i>					
Midline	.50 ^b	-.19	-.17	.13	.18
RH	.59 ^a	-.24	-.19	.18	.26
LH	.35	-.23	-.25	.39	.46

^a $p < .01$.^b $p < .05$.

Friedrich & Friederici, 2006; Holcomb & Neville, 1990; Kutas & Federmeier, 2000; Kutas & Van Petten, 1988; Landi & Perfetti, 2007; Lau et al., 2009; McCleery et al., 2010) the peak amplitudes of the N400 were significantly greater for the semantically incongruent condition than the congruent condition. This 'congruency effect' was significant for the N200 and N400 peaks but not for the N100 peak. Also consistent with previous studies, the N400 congruency effect was particularly prominent in frontal regions (see Fig. 2) (Atchley & Kwasny, 2003 (using linked mastoids reference); Holcomb et al., 1992 (left mastoid reference)).

Some researchers have suggested that early ERP peaks that are sensitive to semantic manipulations may be due to the onset of the N400 (Coulson, Federmeier, Van Petten, & Kutas, 2005). It is unlikely that the N100 response is part of the N200 or N400 responses since it probably had a different source (most prominent in left hemisphere fronto-central channels; Fig. 2) and it was not significantly influenced by the congruency manipulation. It is more likely that the N200 may represent the start of the rise of the N400 component reported in previous studies. This is supported by the finding that these peaks have similar topographies and the difference between congruent and incongruent conditions in these locations becomes more strongly significant for the N400 than the N200 peak (see Fig. 2).

As predicted, there was a moderate correlation between the magnitude of the N400 congruency effect at RH channels and

listening comprehension. However, counter to our predictions, strong and moderate positive correlations were also found between listening comprehension and the congruency effects for the N100 (AFz) and N200 (Cz), respectively. In each case, greater differences in amplitude between congruent and incongruent conditions were associated with better performance on the listening comprehension test. The correlations between listening comprehension and the earlier deflections are surprising because earlier peaks, particularly the N100, most likely reflect sensory rather than cognitive processes. It is possible that these correlations were carried by the very early onset of the N400 for some children who had highly skilled listening comprehension.

These correlations are consistent with Landi and Perfetti (2007) who reported that the N400 effect that occurs in response to picture/word pairs is weaker in adults with less-skilled reading comprehension than adults with skilled reading comprehension. The present data build on the findings of Landi and Perfetti (2007) in two ways. First, we extend their findings to typically developing children and report that children with more skilled listening comprehension show greater congruency effects in the ERP waveform. Second, we show that individual differences in the N400 effect relate specifically to comprehension and not to word recognition or decoding. This validates the N400 effect as a component that captures semantic processing, rather than verbal processing more generally. This finding also provides neurophysiological support

for the view that listening comprehension and decoding have different cognitive and brain bases (Gough & Tunmer, 1986; Keenan, Betjemann, Wadsworth, DeFries, & Olson, 2006; Leonard, Eckert, Given, Virginia, & Eden, 2006) and that listening comprehension depends on a foundation of semantic skills in contrast to word decoding (Bishop & Snowling, 2004). The importance of semantic skills for comprehension is emphasised by the lexical quality hypothesis (Perfetti, 2007) which proposes that rapid and adequate processing of semantic information is necessary for successful comprehension. Based on the present data, it appears that sensitivity to semantic context relates to individual differences in comprehension skills as early as 100 ms post-stimulus onset in children (as evidenced by the significant correlation between comprehension and the N100 effect) and is not restricted to the N400 time window.

There were also correlations between the amplitude of the N100, N200 and N400 peaks (averaged across congruent and incongruent conditions) and non-word decoding, with better decoders showing negative deflections with smaller absolute amplitude. These findings support previous studies that have shown that less-skilled readers generally have more negative ERP waveforms than skilled readers (Landi & Perfetti, 2007; Perfetti et al., 2005; Yang et al., 2005 (using averaged referenced ERPs); Segalowitz et al., 1992 (linked ear referenced ERPs); Meyler & Breznitz, 2005 (using a mastoid reference). We build on these findings by showing that differences in amplitude relate more strongly to non-word decoding skills than to word recognition or comprehension skills. Indeed, there were no significant correlations between word recognition and N100 or N200 averaged amplitudes and only one significant correlation between the averaged N400 amplitudes for LH channels and word recognition. This association between amplitude and phonological decoding skill could reflect an underlying processing difference, as argued by Landi and Perfetti (2007). More specifically, children with better phonological decoding skills may require less effort to process spoken words, hence accounting for their shallower deflections. Conversely, children with less-skilled phonological decoding may process incoming spoken material more intensely than their more skilled peers (Davies, Chang, & Gavin, 2010). This general relationship between amplitude and decoding deserves further investigation, particularly addressing why ERP amplitude relates to non-word reading more strongly than word reading and whether this effect is task specific.

To test the claim that the N400 effect might provide a useful marker of vocabulary knowledge (Byrne et al., 1995a, 1995b, 1999; Connolly et al., 1996; Marchand, D'Arcy, & Connolly, 2002), we investigated whether children's receptive and expressive knowledge of the experimental word stimuli correlated with the magnitude of the ERP effects. We reasoned that children who showed greater sensitivity to the semantic incongruity in ERPs should also be able to provide richer definitions for the same words (if the N400 effect is sensitive to the depth of semantic knowledge) and be more accurate at selecting the correct pictures to go with the words (if the N400 effect is sensitivity to familiarity with the stimuli). However, there were no significant positive correlations between the bespoke vocabulary measures and the incongruity effects for the N100, N200, and more critically, the N400.¹ Since the N400 effect correlated with listening comprehension skill but not vocabulary knowledge, this could suggest that the N400 effect reflects the ease with which semantic information is integrated into context to achieve

comprehension (Brown & Hagoort, 1993; Hagoort, 2008; Osterhout & Holcomb, 1992), rather than reflecting the influence of prior context on access to relevant semantic information (Federmeier, 2007; Kutas & Federmeier, 2000; Lau et al., 2009).

Counter to previous claims, these findings place doubt over the reliability of the N400 effect as a clinical indicator of children's vocabulary knowledge. The lack of correlations between ERP and behavioural measures of vocabulary could be explained by differences in task demands. In contrast to the passive ERP task, the definitions task placed demands on other language skills beyond vocabulary such as pragmatic and syntactic skills, whilst the receptive vocabulary task involved an additional decision element. It is also possible that the correlations between the ERP congruency effects and receptive vocabulary did not reach significance because the scores on this measure were distributed towards ceiling and there was low variability. However, this does not account for the lack of correlations for the definitions task, for which there was good variability in scores.

The present findings highlight the issue that behavioural and neurophysiological methodologies likely tap different (but related) processes. This issue was also raised by Landi and Perfetti (2007) who found a different pattern of group differences (for skilled and less-skilled comprehenders) for the behavioural and ERP data. Such findings are difficult to interpret and we are currently lacking a good understanding of the processes that are captured by each methodology. It is possible that behavioural measures may not be as adequate in revealing subtle differences in the quality of encoded word representations within a group of children as ERP methodology. ERP methodology allows us to examine the very early stages of semantic processing whereas behavioural measures tap the end point of processing. Moreover, behavioural measures often involve conscious introspection of the processes under investigation, in contrast to passive ERP tasks.

An important direction for future research will be to further examine the relationship between behavioural and neurophysiological markers of vocabulary knowledge. This could be achieved by combining the ERP measures with other behavioural indicators (e.g., response time, accuracy) during the ERP task to examine whether the behavioural and neurophysiological data lead to similar conclusions about a child's vocabulary knowledge. Another possibility could be to compare the data obtained from a passive ERP task (as used in the present study) against a number of different behavioural tasks that vary in task demands including off-line (e.g., meaning judgement, definitions) and on-line (e.g., eye tracking, semantic priming) measures, to examine which comparisons lead to the most consistent results. Such studies may lead to increased understanding of brain-behaviour relations and in the longer term to the development of innovative neurocognitive assessment techniques that may be increasingly sensitive to otherwise less noticeable changes.

A limitation of this study is the small sample size. Although power calculations confirmed that the sample size would be large enough to detect N400 effects and strong correlations were obtained with some background measures, a larger sample would have been more ideal to test for relationships between ERP and behavioural measures. It would also be of interest to examine the relationship between neurophysiological and behavioural indices of vocabulary knowledge across a larger age-range, to determine if the weak correlations observed persist across development or rather are dependent on the maturation of strategies.

In conclusion, the present study provides much needed evidence of the N400 effect in middle childhood. Moreover, we have provided evidence, using ERPs, that sensitivity to semantic context relates specifically to listening comprehension, whereas absolute amplitude of negative ERP deflections relates specifically to decoding. These results therefore provide important insights into the

¹ Following Marchand, D'Arcy, and Connolly (2002), we also ran correlations between the measures of vocabulary knowledge and the mean of the *t*-scores obtained from the incongruent and congruent EEG waveform comparison (in addition to the reported Spearman's Rho correlations). None of these correlations were significant.

relationship between component reading skills and associated neural processes and support the view that different foundations of reading (i.e., listening comprehension and non-word decoding) relate differentially to neurophysiological markers (Leonard et al., 2006). The results suggest that semantic processing is fast in children; even from 100 ms after stimulus onset individual differences in sensitivity to semantic context were associated with listening comprehension skill. Finally, the present data suggest that ERP and behavioural indicators of vocabulary knowledge may capture different processes in children and thus, ERPs may not provide a robust estimate of vocabulary knowledge in a clinical setting for the age range studied here. Future research should further investigate these relationships across a larger age and ability range (including children with skilled and non-skilled comprehension and decoding skills) using a wider variety of vocabulary measures.

Acknowledgments

This research was funded by a 1+3 Economic and Social Research Council studentship to the first author. Thanks go to Andre Gouws and Mark Hymers for their help with experiment setup and technical support and G-L Assessment for allowing us to use their stimuli. We would like to thank Nikki Gratton and Charlie Butler

who assisted greatly with data collection and analysis. We would also like to thank the families who participated and the schools who assisted with recruitment. Not least, we would like to thank Charles Perfetti for his valuable discussion and comments.

References

- Atchley, R. A., & Kwasny, K. M. (2003). Using event-related potentials to examine hemispheric differences in semantic processing. *Brain and Cognition*, 53, 133–138.
- Bishop, D. V. M., & Snowling, M. J. (2004). Developmental dyslexia and specific language impairment: Same or different? *Psychological Bulletin*, 130, 858–888.
- Brown, C., & Hagoort, P. (1993). The processing nature of the N400: Evidence from masked priming. *Journal of Cognitive Neuroscience*, 5, 34–44.
- Byrne, J. M., Connolly, J. F., MacLean, S. E., Dooley, J. M., Gordon, K. E., & Beattie, T. L. (1999). Brain activity and language assessment using event-related potentials: Development of a clinical protocol. *Developmental Medicine and Child Neurology*, 41, 740–747.
- Byrne, J. M., Dywam, C. A., & Connolly, J. F. (1995a). An innovative method to assess the receptive vocabulary of children with cerebral palsy using event-related brain potentials. *Journal of Clinical and Experimental Neuropsychology*, 17, 9–19.
- Byrne, M. J., Dywam, C. A., & Connolly, J. F. (1995b). Assessment of children's receptive vocabulary using event-related brain potentials: Development of a clinically valid test. *Child Neuropsychology*, 1, 9–19.
- Coch, D., Maron, L., Wolf, M., & Holcomb, P. J. (2002). Word and picture processing in children: An event-related potential study. *Developmental Neuropsychology*, 22, 373–406.

Appendix A

Congruent (con) and incongruent (incon) word lists.

CON	INCON	CON	INCON	CON	INCON
Package	Palace	Valley	Velvet	Canine	Cavern
Chimney	Cherry	Waistcoat	Warehouse	Pedestrian	Pepperoni
Jewellery	Gentleman	Grooming	Grating	Blazing	Bleeding
Rectangle	Referee	Violin	Volcano	Pillar	Piston
Desk	Deer	Hyena	Hurricane	Wedge	Wealth
Feather	Father	Applauding	Apricot	Polluting	Poverty
Delivering	Decorating	Canoe	Cabin	Foundation	Forestry
Harp	Half	Harvesting	Handkerchief	Clamp	Cliff
Luggage	Lotion	Inflated	Infection	Carpenter	Camouflage
Binoculars	Ballerina	Greeting	Greedy	Departing	Deciding
Funnel	Ferry	Nutritious	Newspaper	Duet	Duty
Calendar	Caravan	Globe	Glove	Illumination	Inoculation
Island	Iron	Hovering	Holiday	Trowel	Trapeze
Links	Lake	Beaker	Beetle	Embracing	Emptying
Gigantic	Gymnastics	Shore	Shed	Citrus	Circuit
Floating	Flicker	Vehicle	Vinegar	Rodent	Roman
Hive	Hill	Currency	Cucumber	Beverage	Boundary
Avocado	Alligator	Hurdling	Horizon	Feline	Furnace
Exhausted	Engagement	Banister	Barnacle	Fungus	Fury
Tearing	Touring	Tubular	Tornado	Interior	Identity
Arctic	Armour	Bouquet	Boulder	Hoisting	Halting
Fountain	Flower	Parallel	Paragraph	Detonation	Dedication
Ankle	Angel	Antlers	Auntie	Consuming	Covering
Chef	Shoe	Escorting	Expansion	Garment	Garage
Sorting	Singing	Aquarium	Academy	Orchard	Orbit
Rough	Run	Hoof	Herb	Pilfering	Pioneer
Tugging	Tunnel	Archaeologist	Accommodation	Primate	Preacher
Target	Tablet	Valve	Vale	Depleted	Devotion
Hook	Hood	Capsules	Cattle	Ascending	Assisting
Sharing	Shaving	Adjustable	Adventurer	Hazardous	Historic
Waterfall	Wandering	Solo	Soda	Goblet	Gossip
Towing	Toasting	Dilapidated	Determination	Appliance	Artistic
Tropical	Triangle	Snarling	Snoozing	Aviation	Apparatus
Fictional	Fisherman	Coast	Court	Talon	Temple
Pelvis	Petal	Construction	Composer		

- Conboy, B. T., & Mills, D. (2006). Two languages, one developing brain: Event-related potentials to words in bilingual toddlers. *Developmental Science*, 9, F1–F12.
- Connolly, J. F., Byrne, J. M., & Dywan, C. A. (1996). Assessing adult receptive vocabulary with event-related potentials: An investigation of cross-modal and cross-form priming. *Journal of Clinical and Experimental Neuropsychology*, 17, 548–565.
- Coulson, S., Federmeier, K. D., Van Petten, C., & Kutas, M. (2005). Right hemisphere sensitivity to word and sentence level context: Evidence from event related brain potentials. *Journal of Experimental Psychology: Learning, memory and Cognition*, 31(1), 129–147.
- D'Arcy, R. C., Connolly, J. F., Service, E., Hawco, C. S., & Houlihan, M. E. (2004). Separating phonological and semantic processing in auditory sentence processing: A high resolution event-related brain potential study. *Human Brain Mapping*, 22, 40–51.
- Davies, P. L., Chang, W. P., & Gavin, W. J. (2010). Middle and late latency ERP components discriminate between adults, typical children and children with sensory processing disorders. *Frontiers in Integrative Neuroscience*, 4(16), 1–9.
- Dunn, L. M., & Dunn, D. M. (1981). *The Peabody Picture Vocabulary Test, Revised*. Minnesota, MN: American Guidance Service.
- Dunn, L. M., & Dunn, D. M. (2009). *The British Picture Vocabulary Scale Third Edition (BPVS-III)*. GL Assessment Limited: London.
- Federmeier, K. D. (2007). Thinking ahead: The role and roots of prediction in language comprehension. *Psychophysiology*, 44, 491–505.
- Fonaryova Key, A. P., Dove, G. O., & Maguire, M. J. (2005). Linking brainwaves to the brain: An ERP primer. *Developmental Neuropsychology*, 27(2), 183–215.
- Friedman, D., Cycowicz, Y. M., & Gaeta, H. (2001). The novelty P3: An event-related brain potential (ERP) sign of the brain's evaluation of novelty. *Neuroscience and Biobehavioural Reviews*, 25, 355–373.
- Friedrich, M., & Friederici, A. D. (2004). N400-like semantic incongruity effect in 19 month olds: Processing known words in picture contexts. *Journal of Cognitive Neuroscience*, 16, 1465–1477.
- Friedrich, M., & Friederici, A. D. (2005). Phonotactic knowledge and lexical-semantic processing in one-year olds: Brain responses to words and nonsense words in picture contexts. *Journal of Cognitive Neuroscience*, 17, 1785–1802.
- Friedrich, M., & Friederici, A. D. (2006). Early N400 development and later language acquisition. *Psychophysiology*, 43, 1–12.
- Gough, P. B., & Tunmer, W. E. (1986). Decoding, reading, and reading disability. *Remedial and Special Education*, 7, 6–10.
- Hagoort, P. (2008). The fractionation of spoken language understanding by measuring electrical and magnetic brain signals. *Philosophical Transactions of the Royal Society B*, 363, 1055–1069.
- Hahne, A., Eckstein, K., & Friederici, A. D. (2004). Brain signatures of syntactic and semantic processes during children's language development. *Journal of Cognitive Neuroscience*, 16(7), 1302–1318.
- Hald, L., Steenbeek-Planting, E. G., & Hagoort, P. (2007). The interaction of discourse context and world knowledge in online sentence comprehension. Evidence from the N400.
- Holcomb, P. J., Coffey, S. A., & Neville, H. J. (1992). Visual and auditory sentence processing: A developmental analysis using event-related brain potentials. *Developmental Neuropsychology*, 8, 203–241.
- Holcomb, P. J., & Neville, H. J. (1990). Auditory and visual semantic priming in lexical decision: Potentials. *Language and Cognitive Processes*, 5, 281–312.
- Keenan, J. M., Betjemann, R. S., Wadsworth, S. J., DeFries, J. C., & Olson, R. K. (2006). Genetic and environmental influences on reading and listening comprehension. *Journal of Research in Reading*, 29, 75–91.
- Kintsch, W., & Rawson, K. A. (2005). Comprehension. In M. J. Snowling & C. Hulme (Eds.), *The science of reading: A handbook* (pp. 209–226). Malden, MA: Blackwell.
- Kutas, M., & Federmeier, K. D. (2000). Electrophysiology reveals semantic memory use in language comprehension. *Trends in Cognitive Sciences*, 12, 463–470.
- Kutas, M., & Van Petten, C. (1988). Event-related brain potentials studies of language. *Advances in Psychophysiology*, 3, 139–187.
- Landi, N., & Perfetti, C. A. (2007). An electrophysiological investigation of semantic and phonological processing in skilled and less skilled comprehenders. *Brain and Language*, 102, 30–45.
- Lau, E., Almeida, D., Hines, P., & Poeppel, D. (2009). A lexical basis for N400 context effects: Evidence from MEG. *Brain and Language*, 111(3), 161–172.
- Lau, E. F., Phillips, C., & Poeppel, D. (2008). A cortical network for semantics: (de)constructing the N400. *Nature Reviews Neuroscience*, 9, 920–933.
- Leonard, C., Eckert, M., Given, B., Virginia, B., & Eden, G. (2006). Individual differences in anatomy predict reading and oral language impairments in children. *Brain*, 129, 3329–3342.
- Luck, S. J. (2005). *An introduction to the event-related potential technique*. Cambridge, MA: MIT Press.
- Marchand, Y., D'Arcy, R. C. N., & Connolly, J. F. (2002). Linking neurophysiological and neuropsychological measures for aphasia assessment. *Clinical Neurophysiology*, 113, 1715–1722.
- McCleery, J., Ceponiene, R., Burner, K. M., Townsend, J., Kinnear, M., & Schreibman, L. (2010). Neural correlates of verbal and nonverbal semantic integration in children with autism spectrum disorders. *Journal of Child Psychology and Psychiatry*, 51(3), 277–286.
- Meyler, A., & Breznitz, Z. (2005). Visual, auditory and cross-modal processing of linguistic and non-linguistic temporal patterns among adult dyslexic readers. *Dyslexia: An International Journal of Research and Practice*, 11(2), 93–115.
- Nanova, P., Lyamova, L., Hadjigeorgieva, M., Kolev, V., & Yordanova, J. (2008). Gender-specific development of auditory information processing in children: An ERP study. *Clinical Neurophysiology*, 119, 1992–2003.
- Nation, K., Marshall, C. M., & Snowling, M. J. (2001). Phonological and semantic contributions to children's picture naming skill: Evidence from children with developmental reading disorders. *Language and Cognitive Processes*, 16(2/3), 241–259.
- Nation, K., & Snowling, M. J. (1997). Assessing reading difficulties: The validity and utility of current measures of reading skill. *British Journal of Educational Psychology*, 67, 359–370.
- Nation, K., & Snowling, M. J. (1998). Semantic processing and the development of word-recognition skills: Evidence from children with reading comprehension difficulties. *Journal of Memory and Language*, 39, 85–101.
- Nation, K., & Snowling, M. J. (1999). Developmental differences in sensitivity to semantic relations among good and poor comprehenders: Evidence from semantic priming. *Cognition*, 70, B1–B13.
- Oldfield, R. C. (1971). The assessment and analysis of handedness: The Edinburgh inventory. *Neuropsychologia*, 9, 97–113.
- Oray, S., Lu, Z. L., & Dawson, M. E. (2002). Modification of sudden onset auditory ERP by involuntary attention to visual stimuli. *International Journal of Psychophysiology*, 43, 213–224.
- Osterhout, L. (2000). On space, time and language: For the next century, timing is (almost) everything. *Brain and Language*, 71, 175–177.
- Osterhout, L., & Holcomb, P. J. (1992). Event-related brain potentials elicited by syntactic anomaly. *Journal of Memory and Language*, 31, 785–806.
- Osterhout, L., McLaughlin, J., & Bersick, M. (1997). Event-related brain potentials and human language. *Trends in Cognitive Sciences*, 1(6), 204–209.
- Perfetti, C. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading*, 11, 357–383.
- Perfetti, C. A., Wlotko, E. W., & Hart, L. A. (2005). Word learning and individual differences in word learning reflected in event-related potentials. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 31(6), 1281–1292.
- Picton, T. W., Bentin, S., Berg, P., Donchin, E., Hillyard, S. A., Johnson, J. R., et al. (2000). Guidelines for using human event-related potentials to study cognition: Recording standards and publication criteria. *Psychophysiology*, 37, 127–152.
- Rodriguez-Fornells, A., Schmitt, B. M., Kutas, M., & Munte, T. F. (2002). Electrophysiological estimates of the time course of semantic and phonological encoding during listening and naming. *Neuropsychologia*, 40, 778–787.
- Schneider, W., Eschman, A., & Zuccolotto, A. (2002). *E-prime user's guide*. Pittsburgh: Psychology Software Tools.
- Segalowitz, S. J., Wagner, W. J., & Menna, R. (1992). Lateral versus frontal ERP predictors of reading skill. *Brain and Cognition*, 20, 85–103.
- Semel, E., Wiig, E. H., & Secord, W. A. (2000). *Clinical evaluation of language fundamentals – third UK edition (CELF – 3)*. The Psychological Corporation: A Harcourt Assessment Company.
- Torgesen, J. K., Wagner, R., & Rashotte, C. (1999). Test of Word Reading Efficiency (TOWRE). Pearson Assessment.
- Torkildsen, J. K., Sannerud, T., Syversen, G., Thormodsen, R., Simonsen, H. G., Moen, I., et al. (2006). Semantic organization of basic-level words in 20-month-olds: An ERP study. *Journal of Neurolinguistics*, 19, 431–454.
- Torkildsen, J. K., Svagstu, J. M., Hansen, H. F., Smith, L., Simonsen, H. G., Moen, I., et al. (2008). Productive vocabulary size predicts event-related potential correlates of fast mapping in 20-month-olds. *Journal of Cognitive Neuroscience*, 20, 1266–1282.
- Vaughan, H. G., Jr., & Ritter, W. (1969). The sources of auditory-evoked responses recorded from the human scalp. *Electroencephalography and Clinical Neurophysiology*, 28, 360–367.
- Vogel, E. K., & Luck, S. J. (2000). The visual N1 component as an index of a discrimination process. *Psychophysiology*, 37, 190–203.
- Wilson, M. (1988). MRC psycholinguistic database—machine-usable dictionary, version 2.00. *Behaviour Research Methods Instruments and Computers*, 20, 6–10.
- Yang, C. L., Perfetti, C. A., & Schmalhofer, F. (2005). Less skilled comprehenders' ERPs show sluggish word-to-text integration processes. *Written Language and Literacy*, 8(2), 157–181.